

Cheng-Tse Liu (Alex)

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EDUCATION

University of California, Santa Cruz

Santa Clara, CA

Sep. 2023 - Dec. 2024

- Master of Science in Natural Language Processing (Computer Science)
- GPA: 3.9/4.0

National University of Kaohsiung

Kaohsiung, Taiwan

Sep. 2018 - Jun. 2022

- Bachelor of Science in Information Management
- GPA: 3.52/4.0, Last 60 credits: 3.8/4.0

TECHNICAL SKILLS

- **Programming Languages & Frameworks** — Python, Java, C, SQL, HTML/CSS, PHP, Flutter, Linux
- **Machine Learning** — PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, Pandas
- **Natural Language Processing** — LLMs, Transformer, RAG, LangChain, Prompt Engineering
- **Data & Cloud Platforms** — MySQL, Firebase, Microsoft Azure, Git

PROFESSIONAL EXPERIENCE

University of California, Santa Cruz

Santa Clara, CA

Feb. 2025 - Present

NLP Software Developer

- Spearheaded new CruzChat features, boosting active-session stability 35% by expanding unit-test coverage and automating GitHub Actions CI pipelines.
- Prototyped Large-Language-Model classroom tools, cutting pilot turnaround from 3 weeks to 5 days through close collaboration with faculty and LangChain/Python rapid-prototyping.
- Diagnosed and Resolved production incidents with < 1-day MTTR, sustaining > 99% uptime via Azure monitoring, log analysis, and hot-fix deployments.

Adobe

Santa Clara, CA

Mar. 2024 - Dec. 2024

Machine Learning Engineer Intern

- Co-designed a multi-agent summarization framework that coordinates GPT-4o, LLaMA-3, and Claude, delivering 2× ROUGE-1 gains vs. single-model baselines on ArXiv & GovReport corpora.
- Implemented a multi-stage chunking + re-summarization pipeline, reducing long-document latency by 42% while preserving factual consistency through dynamic context-window optimization.
- Led prompt-optimization sweeps and built evaluation dashboards that quantified cost-performance trade-offs, trimming Azure OpenAI spend by 28% per experiment.
- Contributed to internal research presentations and conference paper submissions.

Computational Biology & Intelligence System Lab, National University of Kaohsiung

Kaohsiung, Taiwan

Jan. 2021 - Jan. 2022

Research Assistant

- Built a web-scraping and NLP pipeline that mined 200 k+ biomedical abstracts, driving an 83%-F1 BERT classifier for transcription-factor/gene extraction via fine-tuning in PyTorch.
- Benchmarked IRES-prediction tools with ROC analysis and integrated the model into the public Human IRES Atlas database.

PROJECT EXPERIENCE

Multi-LLM Text Summarization

Jan. 2024 - Mar. 2025

- Developed a multi-agent pipeline for efficiently summarizing long documents using state-of-the-art LLMs
- Designed a multi-stage chunking pipeline to process documents beyond standard context limits
- Deployed and evaluated LLMs including GPT-4o, Llama3, Claude, Gemini and Mistral
- Achieved up to 2× performance improvement over single-LLM baselines on ROUGE and BLEU metrics

JeBERTy

Jan. 2024 - Mar. 2024

- Fine-tuned a pre-trained generative model on answer-questioning Japanese dataset
- Trained a BERT model to predicted sentence similarity, reaching 0.89 Cosine Similarity for generated questions on test set

Semantic Textual Relatedness for African and Asian Languages

Sep. 2023 - Jan. 2024

- Fine-tuned BERT model on Hausa named entity recognition, delivering a mean-squared error of 0.03.
- Executed data cleansing, tokenization, and subword feature engineering to extract maximal signal from a limited corpus.

PUBLICATIONS

- Fang, J., **Liu, C. T.**, Kim, J., Bhedaru, Y., Liu, E., Singh, N., ... & Deilamsalehy, H. (2024). Multi-LLM Text Summarization. arXiv preprint arXiv:2412.15487.
- Yang, T. H., Wang, C. Y., Tsai, H. C., Yang, Y. C., & **Liu, C. T.** (2022). YTLR: extracting yeast transcription factor-gene associations from the literature using automated literature readers. Computational and Structural Biotechnology Journal, 20, 4636-4644.
- Yang, T. H., Wang, C. Y., Tsai, H. C., & **Liu, C. T.** (2021). Human IRES Atlas: an integrative platform for studying IRES-driven translational regulation in humans. Database, 2021, baab025.